

1 Vehicle Selection – Joby Aviation

1.1 Concept of Operations

The Joby aircraft is an eVTOL vehicle capable of vertical takeoff and landing, powered by an electric propulsion system, reducing noise and emissions. The aircraft has a passenger capacity of up to 4 passengers, including a pilot. It is primarily designed for short-haul urban air mobility.

Air traffic management systems will incorporate specific routes, altitude restrictions and communication protocols to ensure safe operations. The aircraft is designed to be highly maneuverable in order to maintain safe flight.

Joby aims to provide a seamless and enjoyable passenger experience. Passengers will have access to online booking platforms and mobile applications for flight reservations. In-flight entertainment and communication systems will further enhance the passenger experience.

1.2 Vehicle Requirements / Capabilities

1. Power requirements

- A. The vehicle must be completely electrically powered.
- B. The vehicle must have a backup power supply to ensure safe landing

2. Flight Performance Requirements

- A. The vehicle must have a range of 150 miles.
- B. The vehicle must cruise at a minimum altitude of 7000 feet.
- C. The vehicle must fly at up to 200mph.
- D. The vehicle must land at an H1 helipad.
- E. The vehicle must have a maximum noise level of 65 dB on the ground and while in cruise.
- F. The vehicle must have a pilot making the final decision.
- G. The vehicle must have a self-piloting navigation system.
- H. The vehicle must take off vertically.

3. Sizing Requirements

- A. The vehicle must carry up to four passengers, not including the pilot.
- B. The vehicle must have a maximum takeoff weight of 4800 lbs.
- C. The vehicle must be able to carry a payload of up to 800 lbs,
- D. The vehicle's wingspan and length must fit within an H1 helipad
- E. The vehicle must allow passengers to board without external equipment.

4. Safety Requirements

- A. The vehicle must be capable of continued safe flight in the event of any single failure of the electrical power distribution system.
- B. The vehicle must successfully maneuver through an urban airspace.

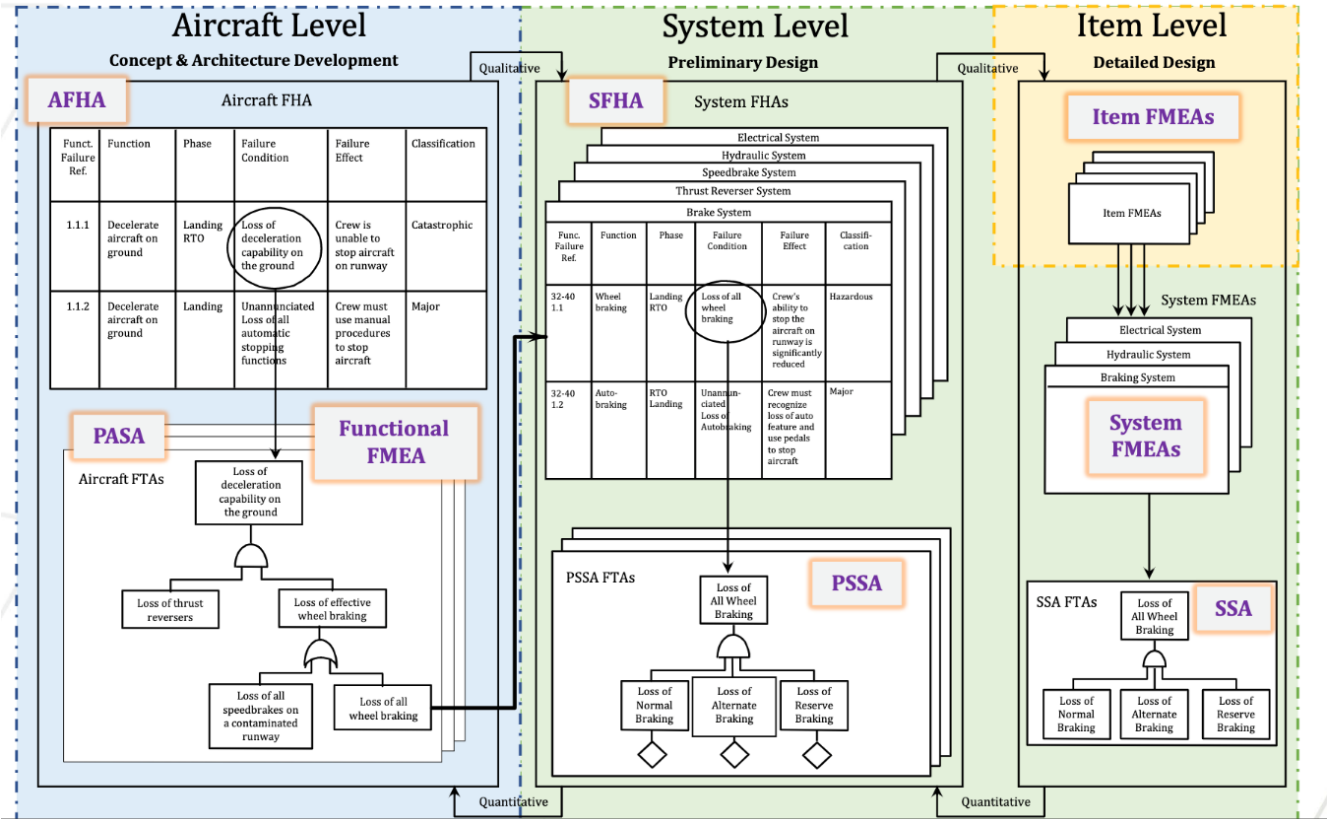
Functional Failure Ref. Phase. Major System. Number

Aircraft Level FHA

Functiona l Failure Ref	Function	Phase	Failure Condition	Failure Effect	Classific ation
1.1.1	Board into vehicle	Taxi	Unable to enter the vehicle	Vehicle	Minor
2.1.1	Generate lift	Takeoff	Unable to generate enough lift to takeoff	Vehicle is unable to take-off the ground	Major
2.1.2	Maintain stable hover	Takeoff/Ho ver	Unable to maintain commanded position	Pilot is unable to maintain control of the aircraft, resulting in either a harsh landing or complete loss of control	Catastro phic

2.1.3	Generate lift	Hover/climb	Unable to generate enough lift to hover/climb	Vehicle is unable to hover/climb above certain altitudes	Major
3.3.1	Tilt actuation	Transition	Tilt motors die on ground	Vehicle fails to forward-transition	Minor
3.3.2	Tilt actuation	Transition	Tilt motors die during cruise	Vehicle fails to backward-transition	Major
3.3.3	Tilt actuation	Transition	Tilt motors die mid-transition	Vehicle gets stuck mid-transition and must land	Hazardous
3.3.4	Tilt state sensing and tilt actuation	Transition	Some tilt motors die, and asymmetry not detected	Vehicle cannot trim and departs controlled flight	Catastrophic
3.3.5	Maintain flight path	Transition	Loss of flight controls	Vehicle departs controlled flight	Catastrophic
4.1.1	Maintain Altitude	Cruise	Unannounced loss of altitude	Crew must glide the aircraft to a controlled crash landing	Hazardous
4.3.1	Control pitch, roll, and or yaw	Cruise	Unannounced loss of flight controls	Crew will have to emergency land wherever.	Catastrophic
4.3.2	Maintain flight path	Cruise	Loss of navigation system	Crew will have to emergency land	Major
4.7.1	Maintain cabin insulation	Cruise	Broken window / open door	Crew will have to slow down and emergency land	Major
5.3.1	Vertical aircraft descent	Landing	Loss of deceleration controls	Pilot is unable to land safely on helipad	Catastrophic
5.4.1	Vertical aircraft descent	Landing	Loss of on-ground navigation	Crew must follow written procedures to decelerate	Major
5.2.1	Vertical aircraft descent	Landing	Unannounced complete loss of energy storage system	Aircraft crashes	Catastrophic
5.2.2	Vertical aircraft descent	Landing	Unannounced partial loss of energy storage system	Pilot descends with on-ground navigation	Major

5.2.3	Vertical aircraft descent	Landing	Failure of regenerative brake systems	Crew is unable to recover and store lost energy	Hazardous
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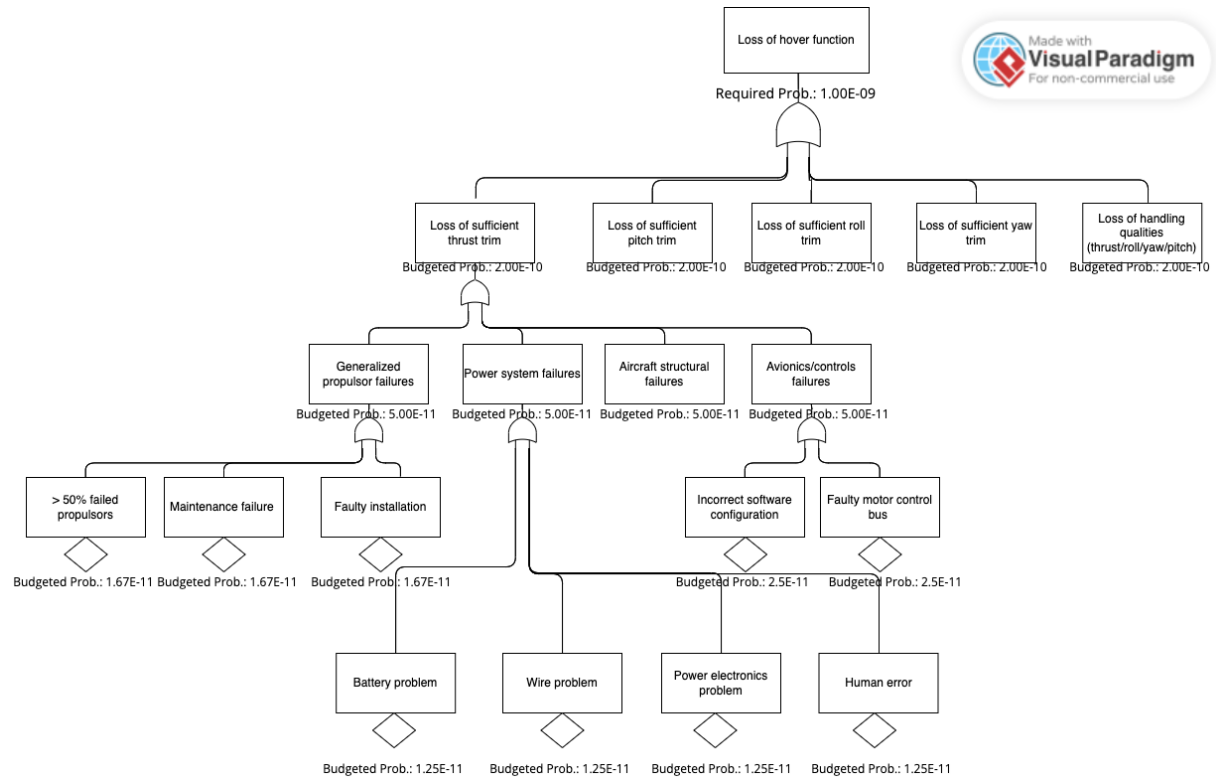


1.3 Fault Tree Analysis?

Loss of hover function - **Required Prob: $<10^{-9}$**

- **Loss of thrust trim**
 - **Generalized Propulsor Failures**
 - **Too many propulsors failed**
 - **Motor failed**
 - **Propeller failed**
 - **Controller failed**
 - **Wire failed**
 - **Maintenance failure**
 - **Faulty installation**
 - **Power System Failure**
 - **Battery problem**
 - **Wire problem**
 - **Power electronics problem**
 - **Crew forgot to charge... someone lost a screwdriver...**
 - **Aircraft Structural Failure**
 - **Something broke or bent**
 - **Avionics/Control Failure**
 - **Software configured incorrectly, shuts off some motors**
 - **Motor control bus not plugged in**
- **Loss of pitch trim**
 - **Generalized Propulsor Losses**
 - **Power System Failure**
 - **Aircraft Structural Failure**
 - **Avionics/Control Failure**
- **Loss of roll trim**
 - **Generalized Propulsor Losses**
 - **Power System Failure**
 - **Aircraft Structural Failure**
 - **Avionics/Control Failure**

- **Loss of yaw trim**
 - **Generalized Propulsor Losses**
 - **Power System Failure**
 - **Aircraft Structural Failure**
 - **Avionics/Control Failure**
- **Degradation of pitch control**
 - **Generalized Propulsor Losses**
 - **Power System Failure**
 - **Aircraft Structural Failure**
 - **Avionics/Control Failure**
- **Degradation of roll control**
 - **Generalized Propulsor Losses**
 - **Power System Failure**
 - **Aircraft Structural Failure**
 - **Avionics/Control Failure**
- **Degradation of thrust control**
 - **Generalized Propulsor Losses**
 - **Power System Failure**
 - **Aircraft Structural Failure**
 - **Avionics/Control Failure**
- **Degradation of yaw control**
 - **Generalized Propulsor Losses**
 - **Power System Failure**
 - **Aircraft Structural Failure**
 - **Avionics/Control Failure**
- **Loss of handling qualities (thrust/roll/pitch/yaw)**
 - **Generalized Propulsor Losses**
 - **Power System Failure**
 - **Aircraft Structural Failure**
 - **Avionics/Control/Input Failure**



System Functional Analysis

Functional Failure Ref	Function	Phase	Failure Condition	Failure Effect	Classification
1.1.1	Generate Thrust	Take-off/Hover	Unable to generate enough thrust for take-off	Propulsor is unable to support take-off	Major
1.1.2	Tilt rotors between hover and forward	Hover/Climb	Unable to tilt the rotors to the correct endpoints	Propulsor fails to tilt between configurations, (causing a significant	Major

	flight configuration			control problem), needing emergency landing.	
1.1.3	Ability to control the pitch of each propellor	Transition	Unable to control the pitch of each propellor	Propulsor fails to generate sufficient thrust, increasing load on other propulsors	Major
1.1.4	Ability to cool system operations	Landing	Unable to cool propulsor down	Propulsor will heat up significantly, potentially causing complete propulsor failure	Hazardous

